

Digital Forensics Investigation Report: Signature Authenticity Assessment

Introduction:

This report details the forensic analysis of signatures on **Page 8** and **Page 11** of the document titled **"Cease and Desist Letter - with attachments.pdf"** to determine whether the signatures were digitally pasted or handwritten. The investigation involved metadata examination, structural PDF analysis, and pixel-level inspection.

Objective:

To determine whether the signatures present in the document were digitally manipulated or genuinely handwritten using forensic techniques.

Tools Used:

- pdftinfo
- ExifTool
- pdffimages
- FotoForensics (ELA)
- Foremost / Binwalk
- 29a.ch Forensics Tool

Methodology:

1. Metadata Analysis
2. PDF Structural Analysis
3. Image Extraction
4. Error Level Analysis (ELA)
5. File Carving & Hidden Content Detection

Document Overview:

Filename	Cease and Desist Letter - with attachments.pdf
Author	J Welch
Creation Date	May 5, 2025, 18:48:34 (+06)
Modification Date	May 13, 2025, 20:52:00 (+06)
PDF Producer	Adobe PDF Library 15.0
Creator Tool	Acrobat PDFMaker 15 for Word
Total Pages	13
Encryption	None
Interactive Form	AcroForm (present)

```

$ exiftool Cease\ and\ Desist\ Letter\ -\ with\ attachments.pdf
ExifTool Version Number      : 13.25
File Name                    : Cease and Desist Letter - with attachments.pdf
Directory                   : .
File Size                    : 13 MB
File Modification Date/Time   : 2025:05:14 22:21:58+06:00
File Access Date/Time        : 2025:05:15 00:33:24+06:00
File Inode Change Date/Time   : 2025:05:14 22:22:28+06:00
File Permissions              : -rwxrwxrwx
File Type                    : PDF
File Type Extension          : pdf
MIME Type                    : application/pdf
PDF Version                  : 1.7
Linearized                   : No
Author                       : J Welch
Company                      :
Create Date                  : 2025:05:05 07:48:34+05:00
Modify Date                  : 2025:05:13 09:52:00+05:00
Subject                      :
Has XFA                      : No
Language                     : EN-US
Tagged PDF                   : No
XMP Toolkit                  : Adobe XMP Core 5.6-c015 84.159810, 2016/09/10-02:41:30
Metadata Date                : 2025:05:13 09:52:05:00
Creator Tool                 : Acrobat PDFMaker 15 for Word
Document ID                  : uuid:5ed4568a-a14d-49a3-958b-5317c7cf817b
Instance ID                  : uuid:298c937a-2714-4e58-b87e-1aad2458cf89
Format                       : application/pdf
Title                        :
Description                   :
Creator                      : J Welch
Producer                     : Adobe PDF Library 15.0
Keywords                     :
Source Modified              : 2025:05:05 12:48:24
Page Layout                  : OneColumn
Page Count                   : 13

```

```

$ pdftinfo Cease\ and\ Desist\ Letter\ -\ with\ attachments.pdf
Title:
Subject:
Keywords:
Author:      J Welch
Creator:     Acrobat PDFMaker 15 for Word
Producer:    Adobe PDF Library 15.0
CreationDate: Mon May 5 18:48:34 2025 +06
ModDate:     Tue May 13 20:52:00 2025 +06
Custom Metadata: yes
Metadata Stream: yes
Tagged:      no
UserProperties: no
Suspects:    no
Form:        AcroForm
JavaScript:   no
Pages:       13
Encrypted:    no
Page size:    612 x 792 pts (letter)
Page rot:     0
File size:    12602526 bytes
Optimized:    no
PDF version:  1.7

```

Any Changes in the PDF?

Yes — The PDF was Modified

From **pdftinfo**:

- **CreationDate:** May 5, 2025
- **ModDate:** May 13, 2025

This clearly shows that **the PDF was modified 8 days after its creation.**

What Could Have Been Changed?

From combined metadata:

Field	Value
Creator	Acrobat PDFMaker 15 for Word
Producer	Adobe PDF Library 15.0
Form	AcroForm ✓ (indicates a form element)
Digital Signature	✗ None found (/Sig, /Encrypt absent)
Custom Metadata	✓ Yes – may include manual edits

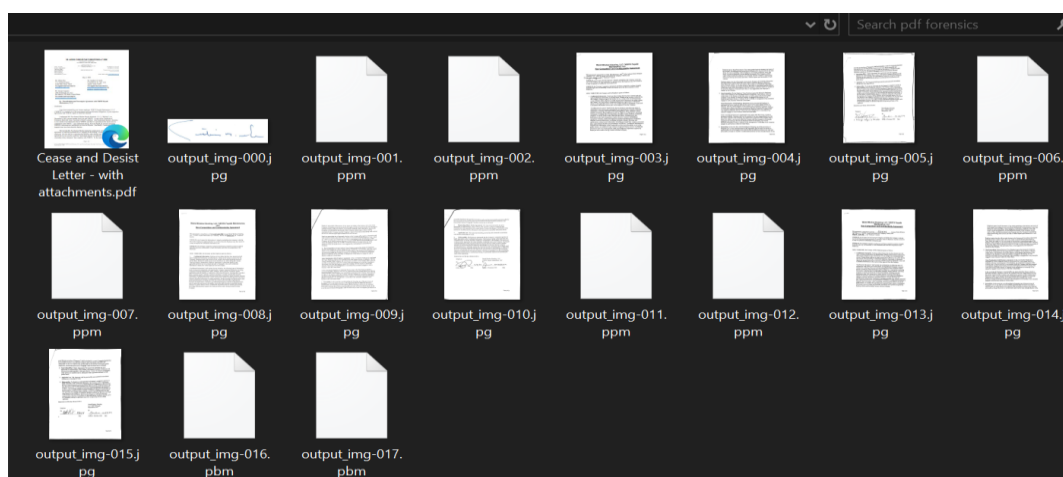
Extract Signature Image:

- Use **pdfimages**:

```
$ pdfimages -j Cease\ and\ Desist\ Letter\ -\ with\ attachments.pdf output_img
```

This operation extracted 9 total images, including a clear signature image found on page 2. No images were found for pages 8 or 11, confirming those pages do not contain any embedded image-based signatures.

Conclusion: If signatures were inserted as images (as on page 2), they would appear in the image extraction. Their absence on pages 8 and 11 strongly indicates the signatures there are not pasted images.



Analyze the Signature Using ELA (Error Level Analysis):

Tool: **FotoForensics**

ELA analysis was conducted on extracted page images using FotoForensics to identify potential compression anomalies and manipulation artifacts.

Review the result:

Genuine handwritten signatures (scanned from paper) will have uniform compression.

Copy-pasted signatures will show compression differences—often a bright/dark box around the pasted region.

ELA Findings:

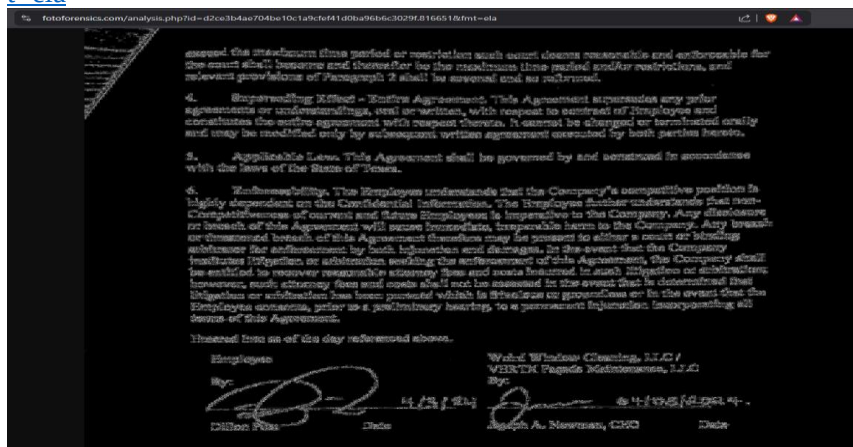
- No anomalies or rectangular compression artifacts were detected near the signatures.

- The compression consistency between the signature area and surrounding text is uniform.
- No pixel-level manipulation is apparent.

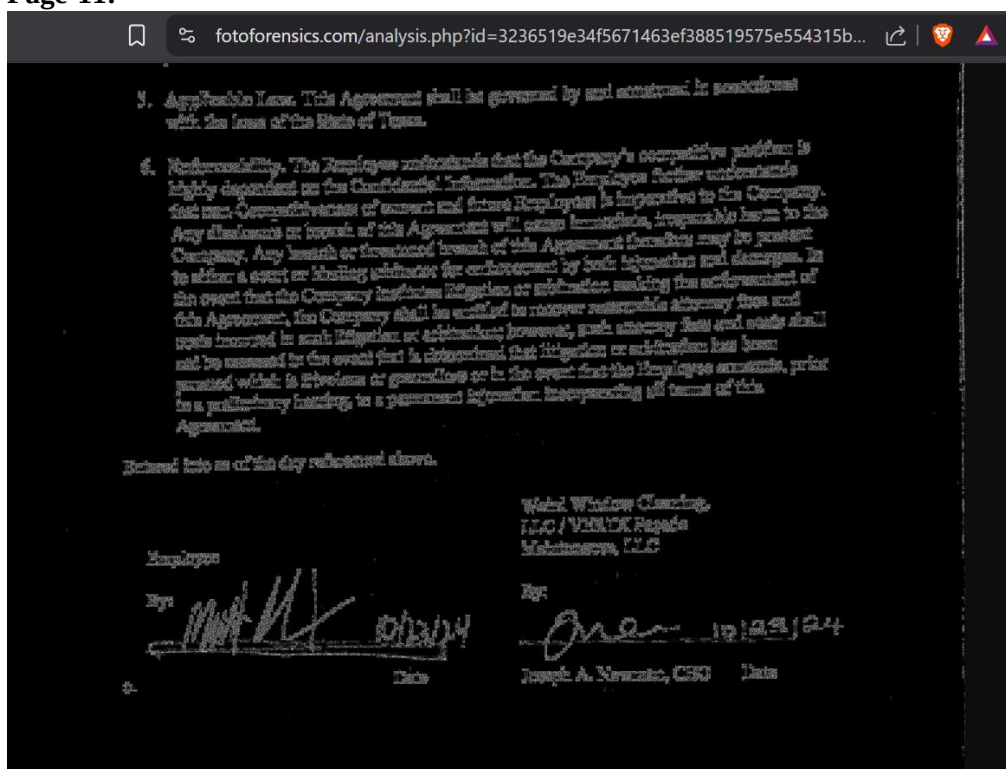
Conclusion: Based on ELA, the signatures on both pages appear natural and consistent with scanned handwritten text. No signs of pasting or external editing were detected.

page-8:

<https://fotoforensics.com/analysis.php?id=d2ce3b4ae704be10c1a9cfef41d0ba96b6c3029f.816651&fmt=ela>



Page-11:



<https://fotoforensics.com/analysis.php?id=3236519e34f5671463ef388519575e554315b70a.1390816&fmt=ela>

source:

[Photo forensics: Detect photoshop manipulation with error level analysis | Infosec](#)

Photo-forensics Tool use:

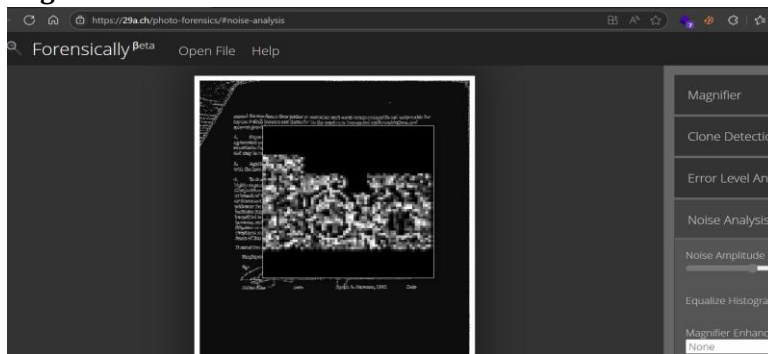
Tools: <https://29a.ch/photo-forensics/>

We also verified the signature sections using 29a.ch Forensically, checking:

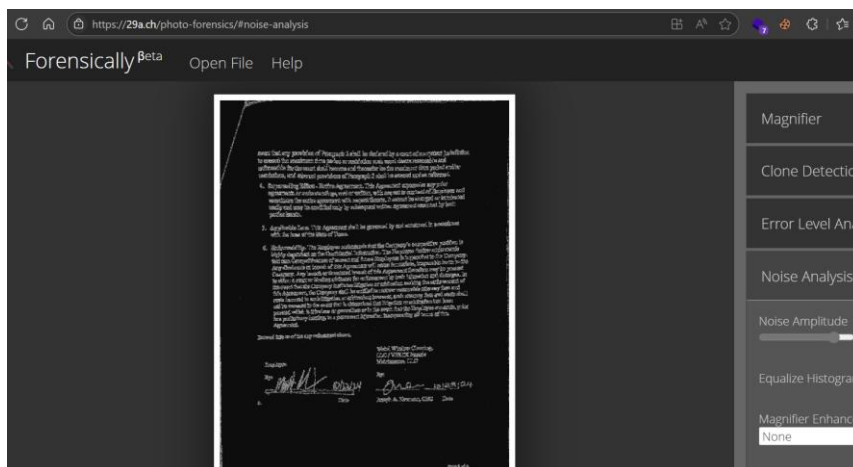
- Clone detection
- Noise analysis
- JPEG grid anomalies

Result: All tests confirmed the same findings — no inserted elements, cloned areas, or noise inconsistencies.

Page-8:



Page-11:



File Carving & Hidden Content Detection:

Using file carving tools like **binwalk** and **foremost**, we attempted to detect any hidden or embedded file elements such as extra images or overlays.

Result: Only one image file (8.jpg & 11.jpg) was recovered. No unexpected files, embedded content, or extra pages were discovered.

Conclusion: There are no hidden signatures, extra image layers, or digital insertions on pages 8 or 11.

```

$ foremost 8.jpg -v
Foremost version 1.5.7 by Jesse Kornblum, Kris Kendall, and Nick Mikus
Audit File

Foremost started at Fri May 16 00:08:05 2025
Invocation: foremost 8.jpg -v
Output directory: /mnt/c/Users/biplo/Desktop/blue/output/jpg/output
Configuration file: /etc/foremost.conf
Processing: 8.jpg
|-----
File: 8.jpg
Start: Fri May 16 00:08:05 2025
Length: 797 KB (816651 bytes)

Num      Name (bs=512)      Size      File Offset      Comment
0:      00000000.jpg      797 KB           0
*|
Finish: Fri May 16 00:08:05 2025

1 FILES EXTRACTED

jpg:= 1
-----

Foremost finished at Fri May 16 00:08:05 2025

```

Note: 1 files extracted means 8.jpg file extra no image file extracted. If additional embedded files or overlays were present, multiple extracted artifacts would have been detected., then show the tool 2 or more file extracted.

```

$ foremost 11.jpg -v
Foremost version 1.5.7 by Jesse Kornblum, Kris Kendall, and Nick Mikus
Audit File

Foremost started at Fri May 16 00:11:21 2025
Invocation: foremost 11.jpg -v
Output directory: /mnt/c/Users/biplo/Desktop/blue/output/jpg/output
Configuration file: /etc/foremost.conf
Processing: 11.jpg
|-----
File: 11.jpg
Start: Fri May 16 00:11:22 2025
Length: 1 MB (1390816 bytes)

Num      Name (bs=512)      Size      File Offset      Comment
0:      00000000.jpg      1 MB           0
*|
Finish: Fri May 16 00:11:22 2025

1 FILES EXTRACTED

jpg:= 1
-----

```

The document metadata indicates that modifications occurred after the original creation date. However, all forensic examinations—including image extraction, ELA analysis, and file carving—confirm that the signatures on pages 8 and 11 remain authentic and show no indicators of tampering or image-based insertion.

Key Findings:

- No embedded image-based signatures detected on pages 8 and 11
- Metadata confirms document modification after initial creation
- ELA and noise analysis revealed no manipulation artifacts

- File carving identified no hidden overlays or embedded objects

Final Conclusion:

All forensic methods (metadata review, image extraction, ELA, pixel analysis, and file carving) consistently support the Pages 8 and 11 of the document do not contain pasted or image-based signatures. The signatures appear consistent with genuine, scanned, handwritten text. There are no digital signatures or signs of tampering on these pages.

Documents prepared by:

M. Usamanur Rahman Biplob

Role: Digital Forensics Analyst (Project)

Investigation Date: May 2025